

RoboNeuro Mapping

Team 47

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Outline

- | **Section 1: Team & Project Info**
- | **Section 2: Methodology**
- | **Section 3: Results & Discussion**
- | **Section 4: Conclusion**
- | **Section 5: Budget & Future Plan**

Team Bio & Member Responsibilities



Qindong Gan

Electrical Engineering

- *Designed the vision embedding module*
- *Prepared presentation materials*



Wentao Zhang

Electrical Engineering

- *Built the core model*
- *Analyzed experiment results and similarity metrics*



Qiyanhui Lu

Electrical Engineering

- *Create the test environments*
- *Contributed to literature review*

Background

The hippocampus and adjacent brain areas are pivotal in the formation of spatial representations that are dependent on context and quickly associating those spaces with reward values.

- **Place Cells:** Fire when the animal is in a specific location in the environment
- **Head Direction Cells:** Encode where the head is facing
- **Boundary Vector Cells (BVCs):** Encode perceived distance to boundaries

Together, they form a **spatial cognitive map** in mammals

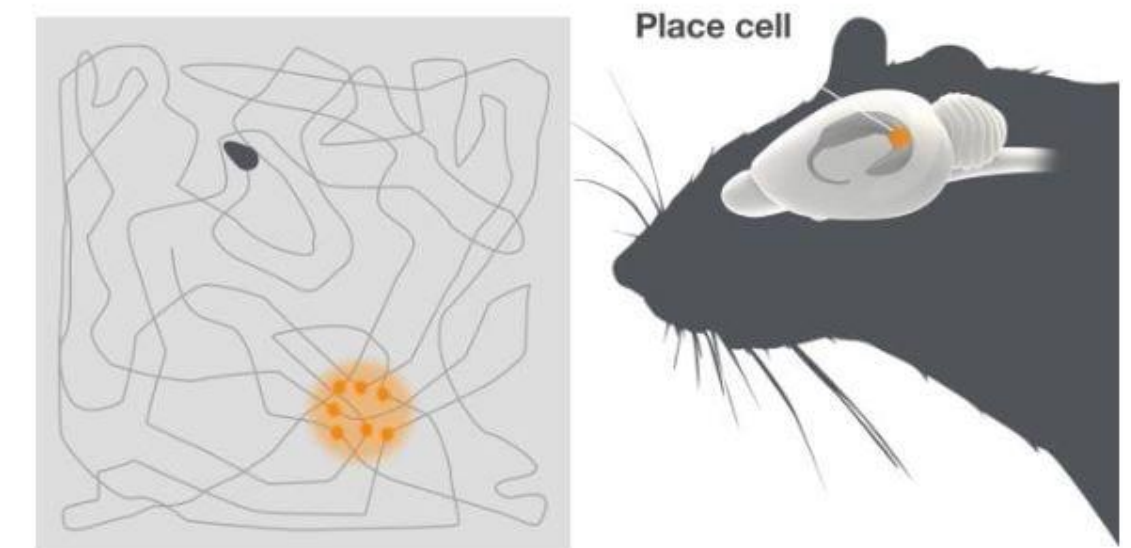


Figure 1. Place Cells

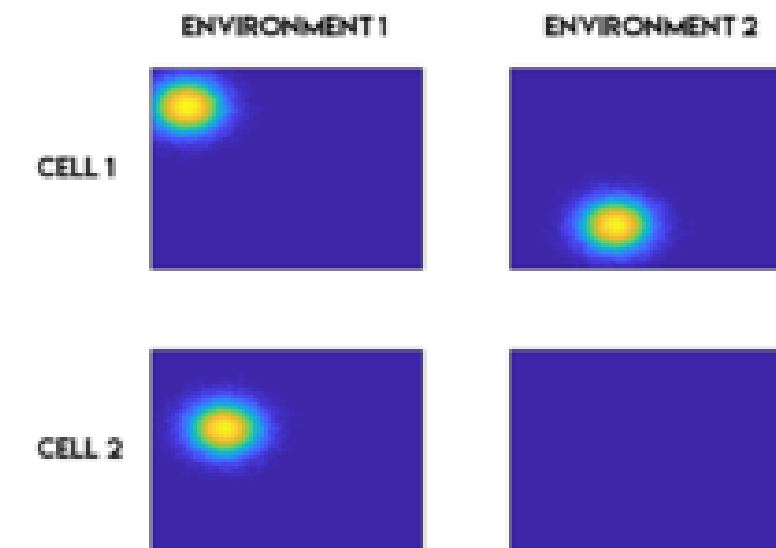


Figure 2. Context Remapping

Background

Previous Limitation

- LiDAR-only models lack semantic awareness [1]
- Hard to disambiguate symmetric or repetitive areas [2]
- Vision provides context → higher-quality place fields

Objective

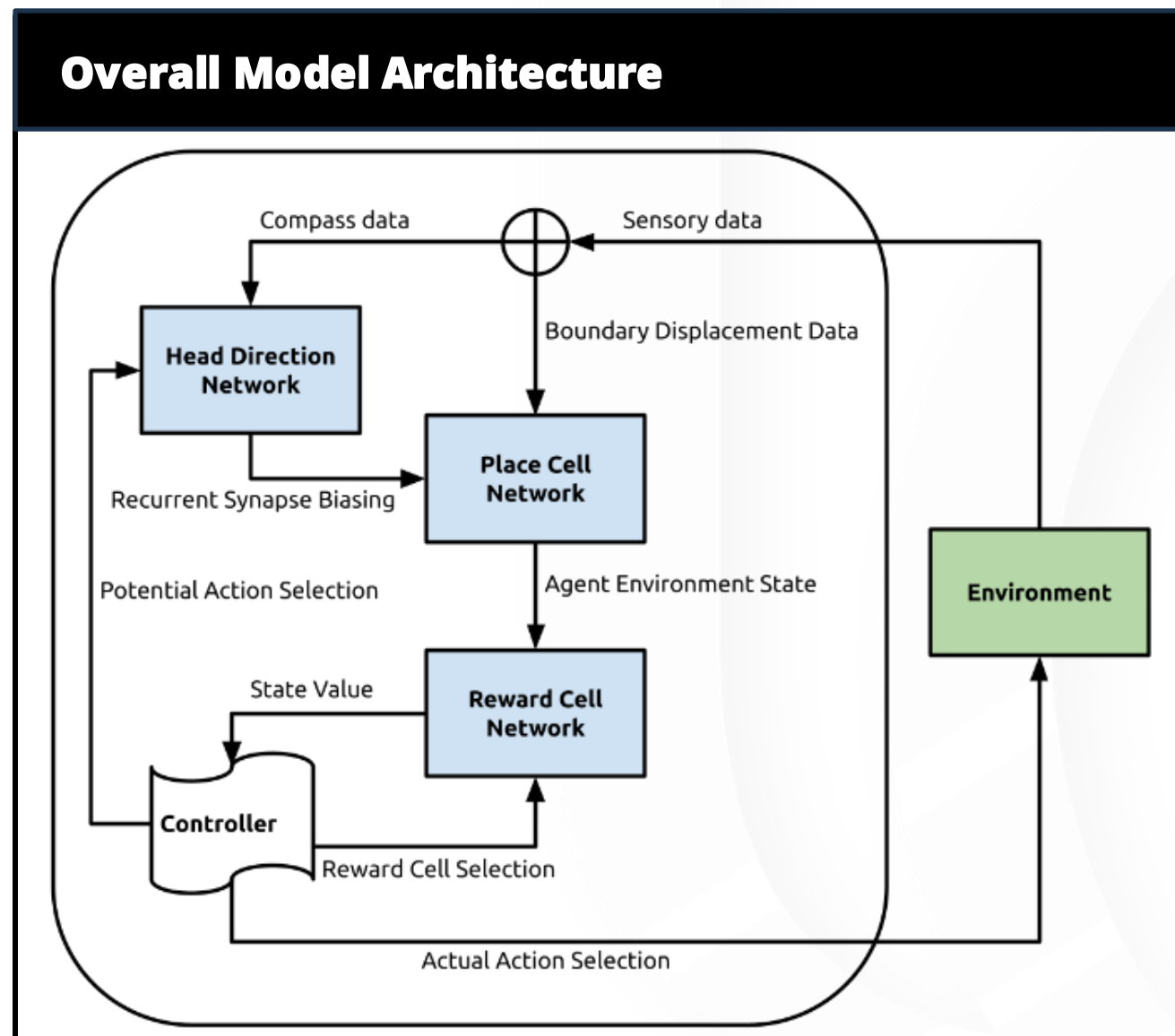
- Enhance place field quality
- Disambiguate visually similar locations [3]
- Improve navigation robustness in ambiguous settings

[1] A. Alabi, A. A. Minai and D. Vanderelst, "One Shot Spatial Learning through Replay in a Hippocampus-Inspired Reinforcement Learning Model," 2020 International Joint Conference on Neural Networks (IJCNN), Glasgow, UK, 2020, pp. 1-8, doi: 10.1109/IJCNN48605.2020.9207435

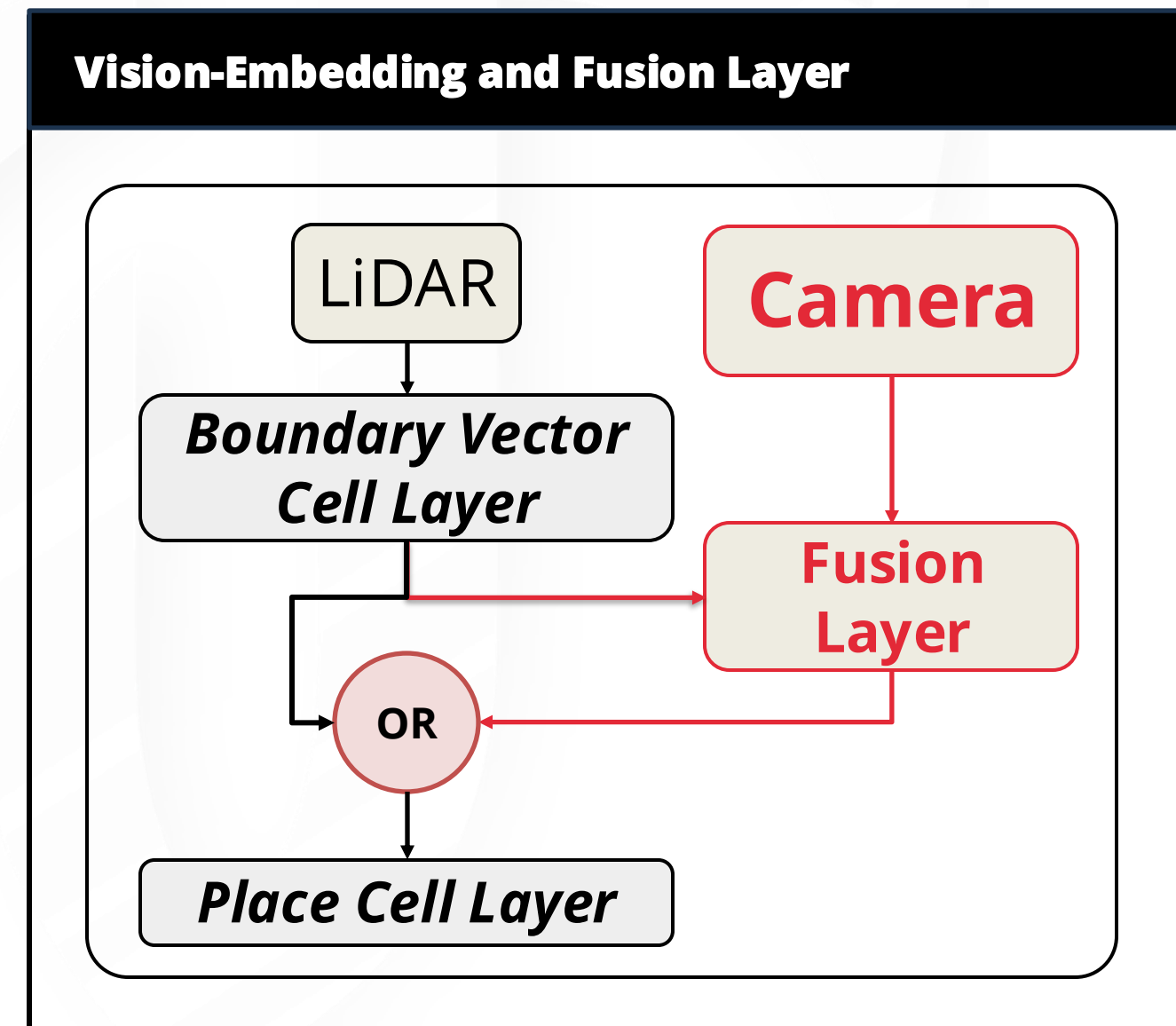
[2] A. Alabi, D. Vanderelst and A. A. Minai, "Context-Dependent Spatial Representations in the Hippocampus using Place Cell Dendritic Computation," 2022 International Joint Conference on Neural Networks (IJCNN), Padua, Italy, 2022, pp. 1-8, doi: 10.1109/IJCNN55064.2022.9892401.

[3] Alabi A, Vanderelst D, Minai A A. Rapid learning of spatial representations for goal-directed navigation based on a novel model of hippocampal place fields[J]. Neural Networks, 2023, 161: 116-128.

Proposed Solution



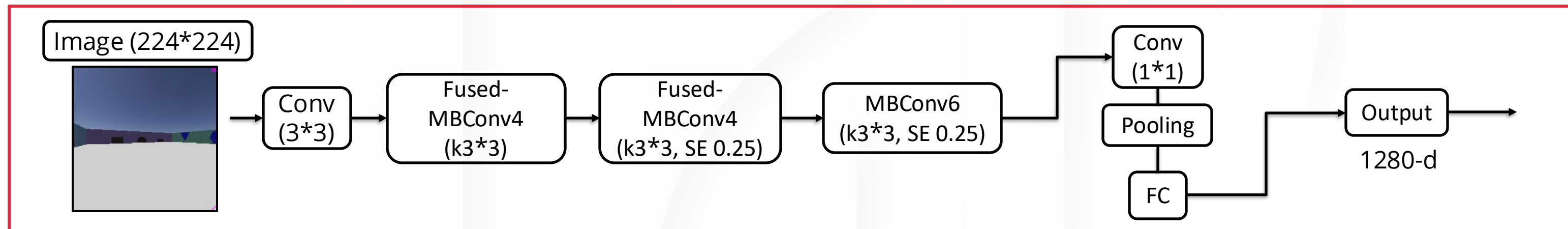
A bio-inspired reinforcement learning architecture integrating head direction, place cell, and reward cell networks for autonomous navigation.



The red section represents our vision embedding model, which fuses BVC outputs and camera inputs before feeding them to the place cell layer.

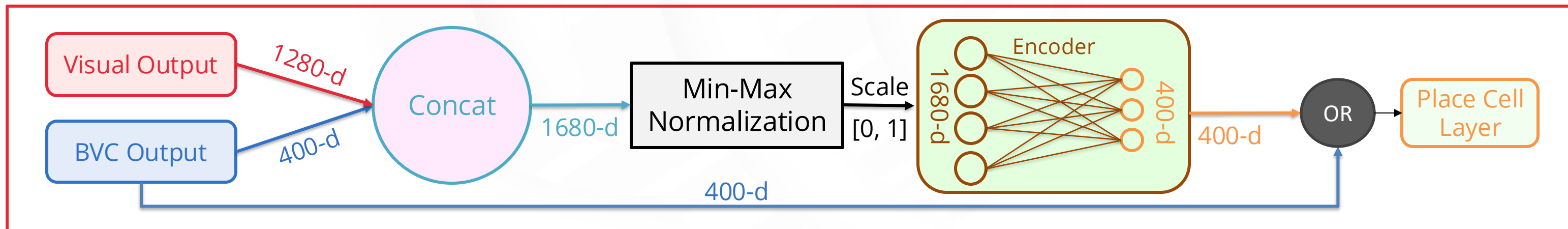
Proposed Solution

Vision-Embedding



EfficientNetV2 Model Architecture Diagram.

Fusion Layer



BVC and Visual outputs are passed through a fusion layer to generate a 400-dimensional fused feature vector.

Design Alternatives & Standards Compliance

EfficientNetV2-S

```
# about base * (width1.4, depth1.8)
v2_s_block = [
    'r2_k3_s1_e1_i24_o24_c1',
    'r4_k3_s2_e4_i24_o48_c1',
    'r4_k3_s2_e4_i48_o64_c1',
    'r6_k3_s2_e4_i64_o128_se0.25',
    'r9_k3_s1_e6_i128_o160_se0.25',
    'r15_k3_s2_e6_i160_o256_se0.25',
]
```

EfficientNetV2-M

```
# about base * (width1.6, depth2.2)
v2_m_block = [
    'r3_k3_s1_e1_i24_o24_c1',
    'r5_k3_s2_e4_i24_o48_c1',
    'r5_k3_s2_e4_i48_o80_c1',
    'r7_k3_s2_e4_i80_o160_se0.25',
    'r14_k3_s1_e6_i160_o176_se0.25',
    'r18_k3_s2_e6_i176_o304_se0.25',
    'r5_k3_s1_e6_i304_o512_se0.25',
]
```

EfficientNetV2-L

```
# about base * (width2.0, depth3.1)
v2_l_block = [
    'r4_k3_s1_e1_i32_o32_c1',
    'r7_k3_s2_e4_i32_o64_c1',
    'r7_k3_s2_e4_i64_o96_c1',
    'r10_k3_s2_e4_i96_o192_se0.25',
    'r19_k3_s1_e6_i192_o224_se0.25',
    'r25_k3_s2_e6_i224_o384_se0.25',
    'r7_k3_s1_e6_i384_o640_se0.25',
]
```

Mobile Devices
Embedded Devices

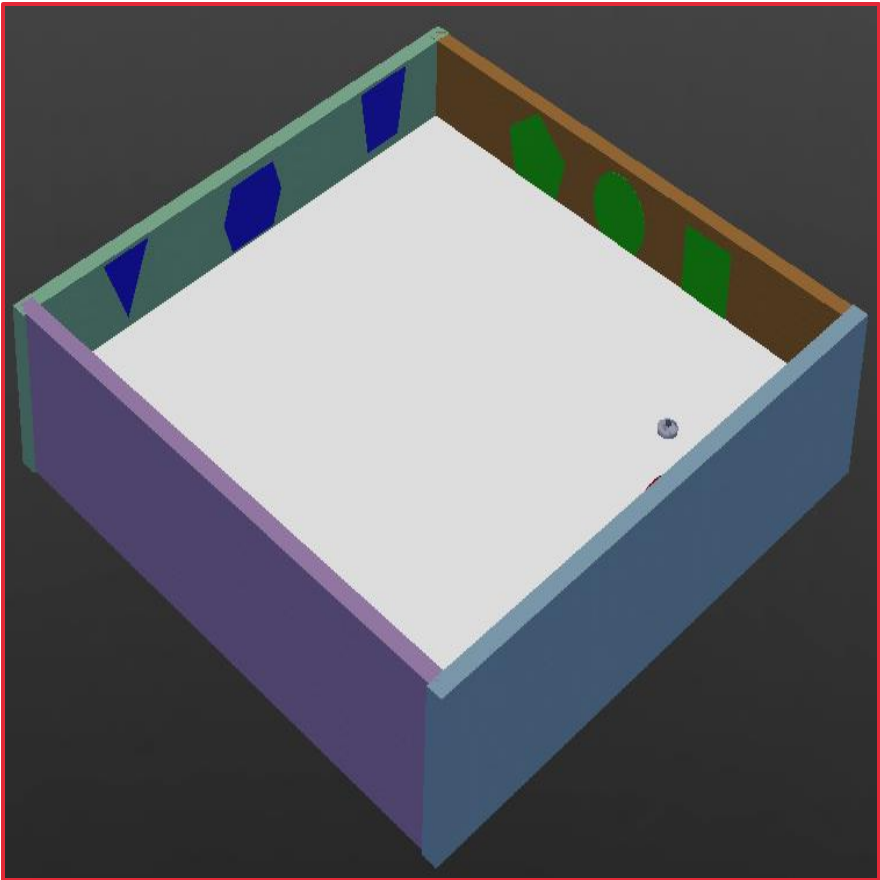
Medium-Scale
Servers

High-
Performance
GPU Servers

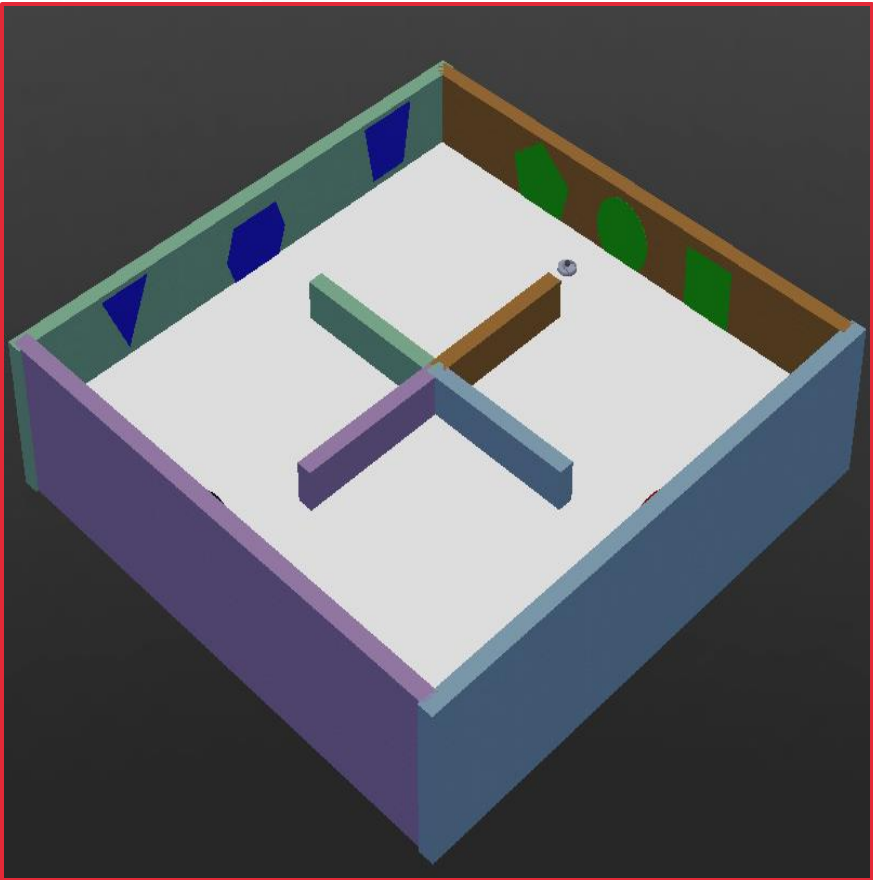
Depending on task requirements

- PEP 8 – Python Style Guide:** All Python code (in core/, controllers/, visualizations/) conforms to PEP 8.
- Git Version Control & Branching:** We use a GitFlow branching model: feature branches, pull requests, code reviews, and semantic version tags.
- IEEE 1872 – Ontologies for Robotics & Automation:** Our neural layers (place cells, reward cells, head-direction cells) can be mapped to standard robotic ontologies for future semantic interoperability.
- MIT License:** We declare an open-source MIT license in our Github Repo:
https://github.com/bduke13/AlabiHippocampalModel/tree/senior_design_vision.

Experimental Settings



Easy Environment



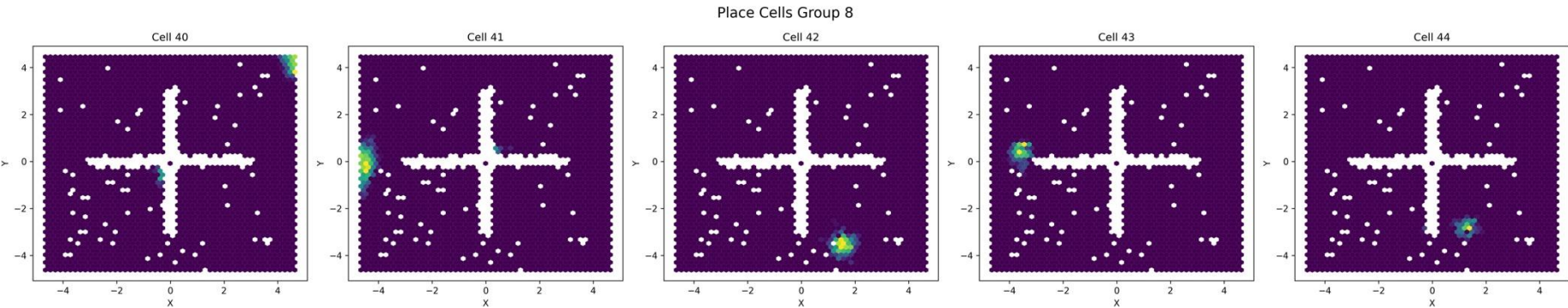
Complex Environment

Parameter	Value	Description
Place Cells	500	Number of neurons in the place cell network.
BVCs	50	Number of BVCs per direction.
Head Direction	8	Number of head direction cells.
Time Step (ms)	96	Duration of each simulation step (32 × 3).
Radial Sigma	0.5	Standard deviation for the radial component.

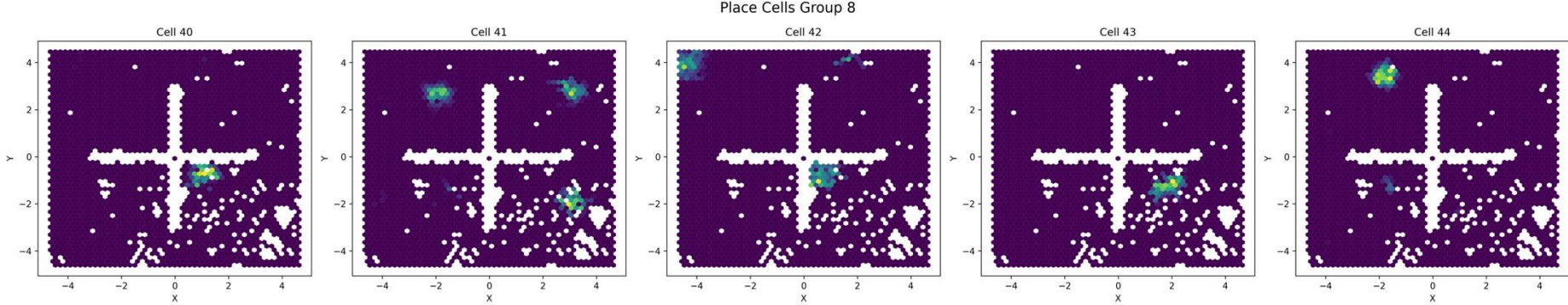
The test environment employs a 10 × 10 world, with each of the four walls featuring a unique color and pattern. In more complex settings, we additionally place a cross in the center to introduce interference for the robot.

Final Results

Enable Vision-Embedding

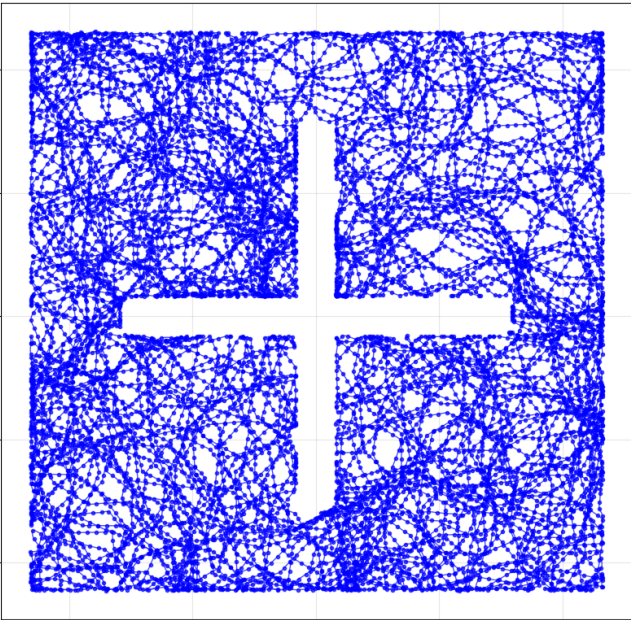


Disable Vision-Embedding

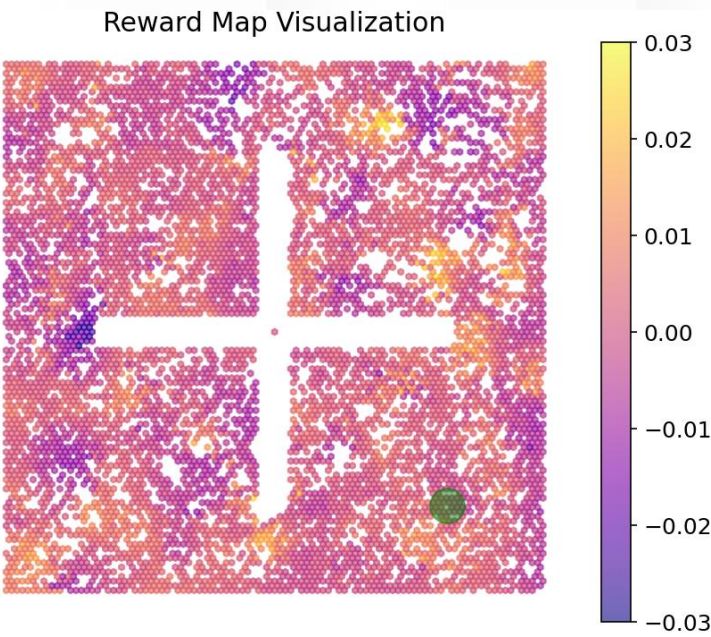


Place cell field Group
more focused and brighter, with greater distinction

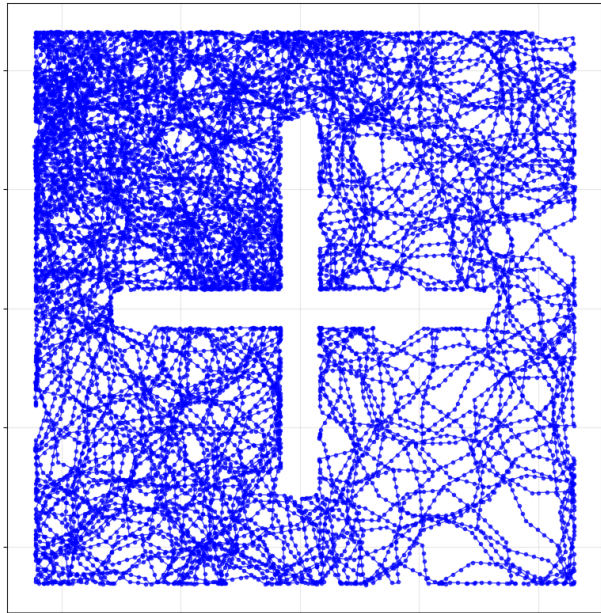
Place cell field Group
overlapping, dispersed, and uncertain



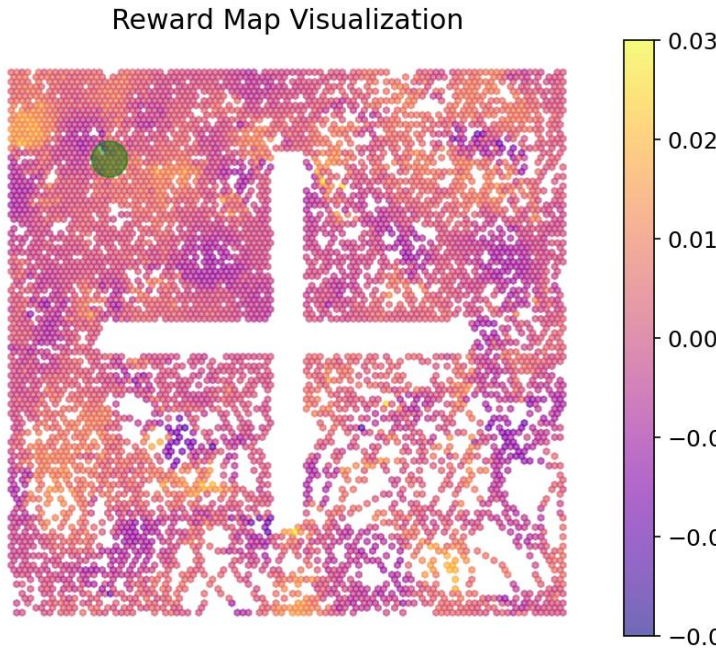
Agent Path



Reward Map



Agent Path



Reward Map

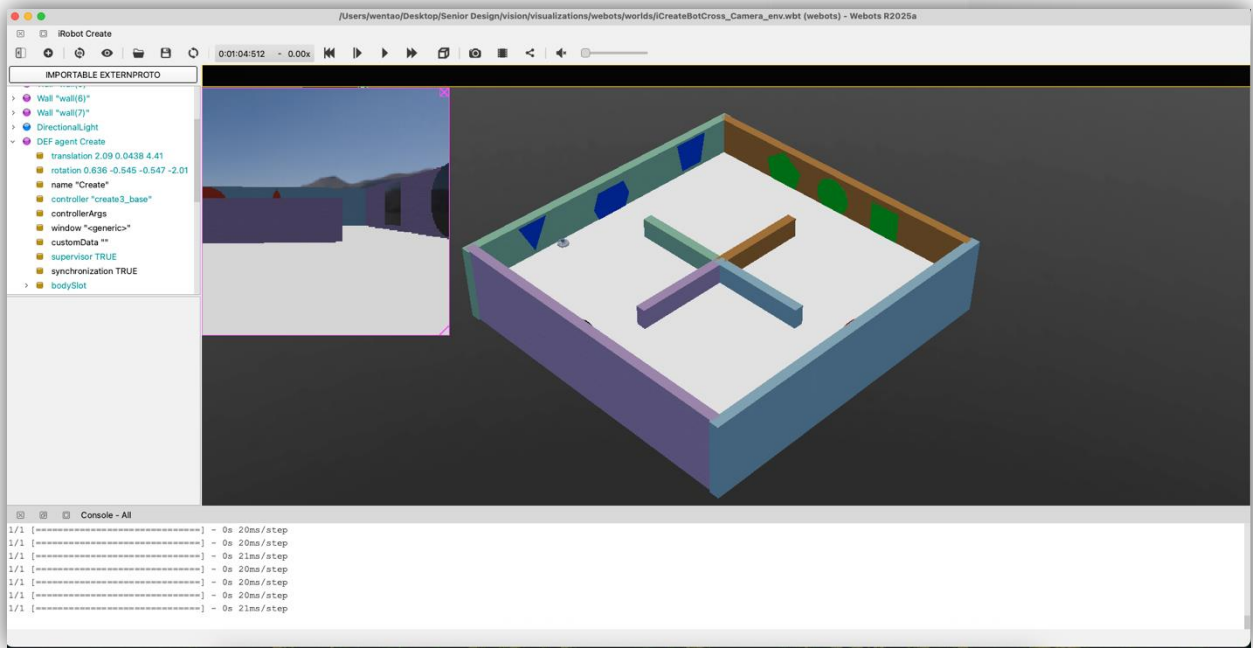
Conclusion

In conclusion, the value of this work lies in its systematic exploration of the boundaries of visual fusion in neural navigation.

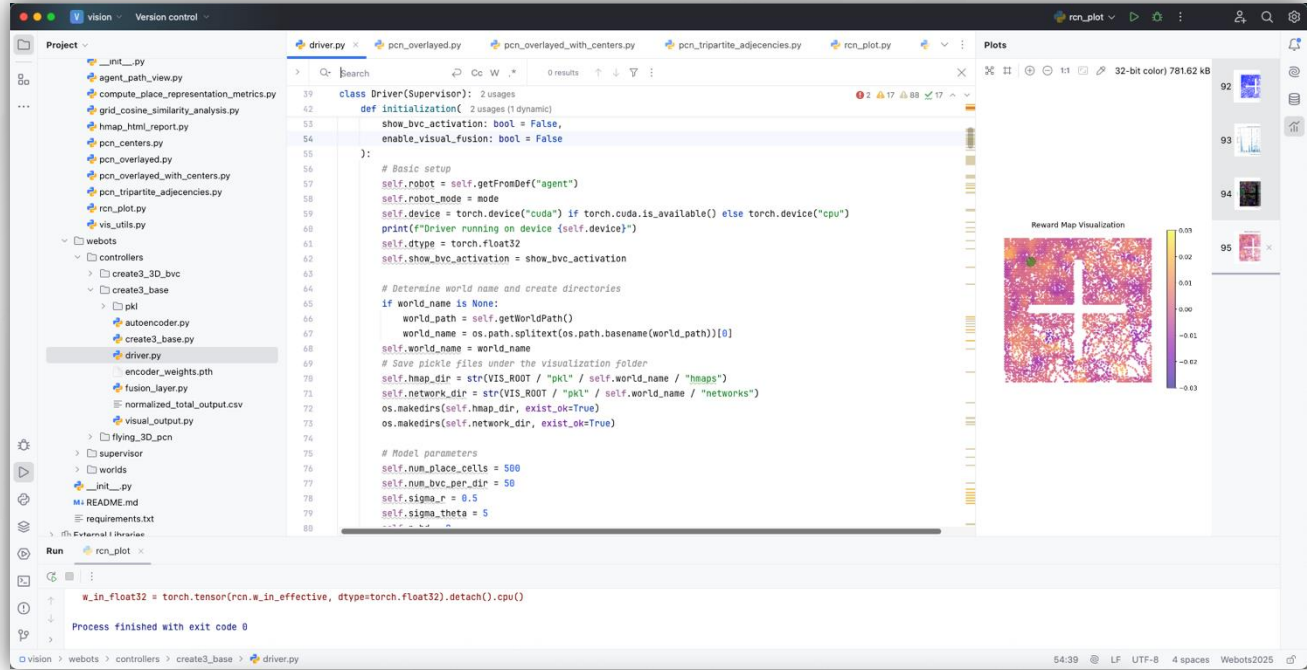
With the addition of visual input, the model leveraged wall colors and patterns as effective spatial cues, enabling the robot to accurately localize itself and avoid getting stuck in a corner. Our results indicate that visual features serve better as ***semantic enhancers*** than as ***primary spatial encoders***, complementing LiDAR's strength in geometric resolution.

While visual input holds promise for enriching spatial perception, its effectiveness depends heavily on the architectural design, learning paradigm, and fusion methodology.

Budget



Webots R2025Platform



Python Control Code

The Total Cost is
0\$

Design & Optimization

Hardware Deployment

Algorithm



Physical Robot

What we are doing
Most in Software Simulation

- 1. Reasonable budget savings
- 2. Appropriate time planning

Challenges & Future Plans

Challenges

1. Spatial Sensitivity
Low Spatial Sensitivity
of Vision Model

2. Fusion
Limited Fusion
Effectiveness

3. Generalization
Poor Generalization in
Complex Scenes

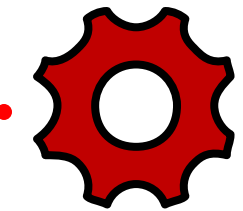
4. Geometric Alignment
Weak Cross-Modal
Alignment

Future Plans



Use Spatially-Aware Vision Models

Replace EfficientNetV2 with models like **ViT** or **CoordConv** for better spatial encoding.



Adopt Multi-View Image Encoding

Capture **viewpoint-dependent** features to **reduce ambiguity**.



Apply Cross-Modal Contrastive Learning

Align **visual** and **LiDAR** features using the **contrastive loss** and **attention-based fusion**.

Reference

1. Alabi, A., Vanderelst, D., and Minai, A. A. "Rapid Learning of Spatial Representations for Goal-Directed Navigation Based on a Novel Model of Hippocampal Place Fields." *Neural Networks*, vol. 161, 2023, pp. 116–128.
2. Alabi, A., Vanderelst, D., and Minai, A. A. "Context-Dependent Spatial Representations in the Hippocampus Using Place Cell Dendritic Computation." *Proceedings of the IEEE World Congress on Computational Intelligence (WCCI)*, Padua, Italy, July 2022, pp. 1–10.
3. Gerstenslager, A., and Dukenbaev, B. "Using a Simulated Place-Cell Model for Autonomous Reward Mapping and Navigation on a Robotic Platform." Bachelor's thesis, Department of Electrical Engineering and Computer Science, University of Cincinnati, Apr. 2024.
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11. Pandey, Krishna Kant, Prases K. Mohanty, and Dayal R. Parhi. "Real Time Navigation Strategies for Webots Using Fuzzy Controller." *2014 IEEE 8th International Conference on Intelligent Systems and Control (ISCO)*, IEEE, 2014.

Thank You!